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Enrich and Detect: Video Temporal Grounding with Multimodal LLMs

Shraman Pramanick[†1,2✉] Effrosyni Mavroudi[1] Yale Song[1] Rama Chellappa[2]
Lorenzo Torresani[◇1] Triantafyllos Afouras[1✉]
[1]FAIR, Meta [2]Johns Hopkins University [3]Northeastern University

Abstract

We introduce ED-VTG, a method for fine-grained video temporal grounding utilizing multi-modal large language models. Our approach harnesses the capabilities of multimodal LLMs to jointly process text and video, in order to effectively localize natural language queries in videos through a two-stage process. Rather than being directly grounded, language queries are initially transformed into enriched sentences that incorporate missing details and cues to aid in grounding. In the second stage, these enriched queries are grounded, using a lightweight decoder, which specializes at predicting accurate boundaries conditioned on contextualized representations of the enriched queries. To mitigate noise and reduce the impact of hallucinations, our model is trained with a multiple-instance-learning objective that dynamically selects the optimal version of the query for each training sample. We demonstrate state-of-the-art results across various benchmarks in temporal video grounding and paragraph grounding settings. Experiments reveal that our method significantly outperforms all previously proposed LLM-based temporal grounding approaches and is either superior or comparable to specialized models, while maintaining a clear advantage against them in zero-shot evaluation scenarios.

1. Introduction

Video temporal grounding [32, 40, 53, 80, 102] aims to identify temporal intervals in a video that correspond to a set of provided language queries. The task is essential for applications such as video editing and content retrieval. Conversely, video captioning [1, 59, 78, 79, 104] entails generating a natural language description for a given video segment, effectively translating visual content into text. The two tasks are in fact dual, as the outputs of one task are the inputs to the other, and vice versa. Intuitively, there is significant potential in exploiting this synergy, however it has largely remained unexplored, with previous works typically specializing in one of the two tasks [8, 32, 53, 104] or solving them in multi-task setting [72, 85] without investigating how each task can benefit the other.

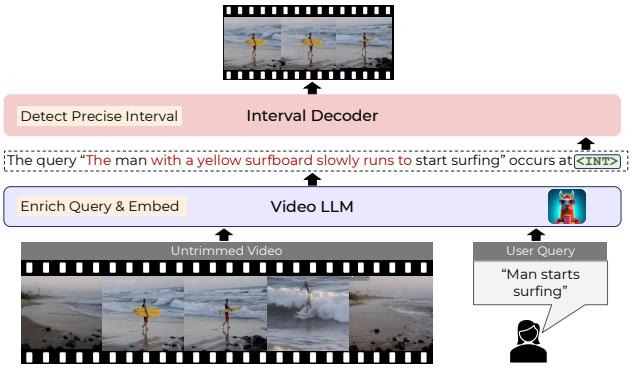

Figure 1. **Our proposed system.** ED-VTG performs video temporal grounding as a two-stage process: since user queries are often incomplete or coarse, the first stage involves producing an enriched query that adds additional details to the original, making it easier to ground. Meanwhile, a contextualized embedding is generated, containing all the information about the interval to be predicted. In the second stage, an interval decoder translates these embeddings into precise temporal boundaries.

In this work, we exploit this duality by leveraging captioning to enhance grounding. Our key observation is that natural language queries often lack the completeness or detail necessary for effective temporal localization; indeed, existing grounding datasets frequently contain poorly worded, coarse, and potentially incomplete queries. The quality and completeness of these queries however is crucial for the precision of the temporal grounding. A natural hypothesis, therefore, is that more detailed queries, which could be obtained via conditional captioning, can significantly enhance grounding accuracy. For example, as illustrated in Figure 1, a vague query like *'Man starts surfing'* can be refined into a more detailed description such as *'The man with a yellow surfboard slowly runs to start surfing,'* resulting in grounding with more accurate temporal boundaries. In other cases, refining an abstract concept may involve breaking down a complex query into simpler components that are more directly groundable.

This key idea of query enrichment forms the basis of our proposed approach. Concretely, we transform grounding into a two-stage reasoning process using a multi-modal

[†]Work done during an internship at FAIR.
[◇]Work done while at FAIR.
[✉]spraman3@meta.com, afourast@meta.com

LLM: first the model enriches the input query into a more detailed description by adding missing details based on the video content, and then temporally localizes the resulting enriched query in the video.

To effectively perform the temporal localization, we introduce a lightweight perception decoder that specializes in generating precise temporal boundaries, conditioned on a contextual representation, allowing the LLM to focus on language outputs, where it excels. The perception decoder allows us to leverage purposefully crafted training objectives for temporal grounding, building on prior knowledge and task-specific characteristics developed from years of research in object detection [17, 19, 30, 67, 71, 77] and temporal localization [32, 40, 53, 61, 102, 105].

Learning to jointly *enrich and detect* requires high-quality enriched query labels. We obtain those by using a strong external captioning model which we condition on the original queries and the video content of the target temporal boundaries. However powerful, these models are prone to hallucinations and there is no guarantee that the enriched queries will always be easier to ground than the original ones. At the same time, annotating the ground truth to determine which query – original or enriched – is a better candidate, is extremely expensive. To address this issue, we propose training in a multiple-instance learning (MIL) fashion, that enables the model to autonomously determine which query is better suited for the task during training.

Finally, we note that our proposed method is not equivalent to a data augmentation approach which simply preproecsses the training set to generate enhanced queries that are directly used for training. While this simpler alternative offers some of the same benefits, it suffers from the limitation that extracting enriched queries during training requires knowledge of the ground-truth temporal segments, *i.e.*, the grounding targets. Since during inference these segments are unknown, the original queries must be used as input, which, as we will show experimentally, is suboptimal. Our method overcomes this limitation by learning to jointly enrich and detect, demonstrating superior performance.

To summarize, our contributions are as follows: (i) we introduce a cascaded approach to temporal grounding, where the model first enriches the provided language query based on the video context and then proceeds to localize it; (ii) we enable multi-modal LLMs to accurately localize text queries using a lightweight decoder which allows for training with detection objectives tailored to the task; (iii) we propose a multiple-instance learning paradigm that enables the model to dynamically select the query that leads to better temporal localization; (iv) we achieve state-of-the-art results on several temporal grounding benchmarks, for both single query grounding and paragraph grounding, demonstrating, for the first time, an LLM-based model that surpasses or performs comparably to specialist models.

2. Related Works

LLM-based temporal grounding. Prior works have explored using LLMs for grounding natural language sentences in videos, either using raw text tokens to represent timestamps [21, 36, 39, 48, 72, 99] or by adding hundreds of special tokens to the LLM's vocabulary to represent video frames [22, 63, 85]. Our approach differs from these methods in that by utilizing a lightweight interval decoder we can apply detection losses such as L1 and gIoU with minimal added complexity; we additionally exploit the LLM's potential to describe video content in detail.

Specialist models. There is a rich variety of specialist models in the literature that are tailored to specific variants of temporal grounding, *e.g.* single query temporal grounding [102] and video paragraph grounding [82], and as such, achieve strong performance [6, 26, 41, 46, 75, 82, 98, 105, 106]. Modern methods typically employ a multi-modal transformer [102] that fuses dense video features with text embeddings of the language query, followed by a specialized detector head for performing detections [16, 32, 53, 74–76]. However, because these models are often trained on limited datasets for such a narrowly defined task, they struggle with generalization. Indeed, the zero-shot performance of these methods is limited [13, 89, 90, 92, 94]. In this work, we aim to address the shortcomings of previous methods by fully combining the generalization abilities of multi-modal LLMs with the advantages of specialist models.

Dense captioning. Our method is related to dense video captioning [29], where the objective is to segment a given video into multiple parts and simultaneously provide descriptive captions for each segment. Traditional approaches tackle this task either by first determining the segments and then providing descriptions [6, 21, 25, 38, 111, 113], or jointly learning both tasks [10, 38, 93, 112, 116]. Recent advances have demonstrated video-conditioned LLMs to excel in this task [88, 115, 117]. While there are similarities between our enrich-and-detect paradigm and dense captioning, we solve a different task, namely video temporal grounding, where the input query is given and constrains the problem.

Prompt augmentation with LLMs. Off-the-shelf LLMs have recently been used to augment input prompts for tasks such as image retrieval and image classification by providing additional, clarifying descriptions. These augmented descriptions can aid generalization in various vision and NLP tasks such as visual question answering [12, 62, 84], dialog generation [15], and visual classification [51]. Our approach goes beyond using an off-the-shelf LLM for offline query enrichment, rather learning to dynamically enrich queries during inference time towards better grounding.

Prior work has also investigated the use of captioners and language models for data augmentation by rephras-

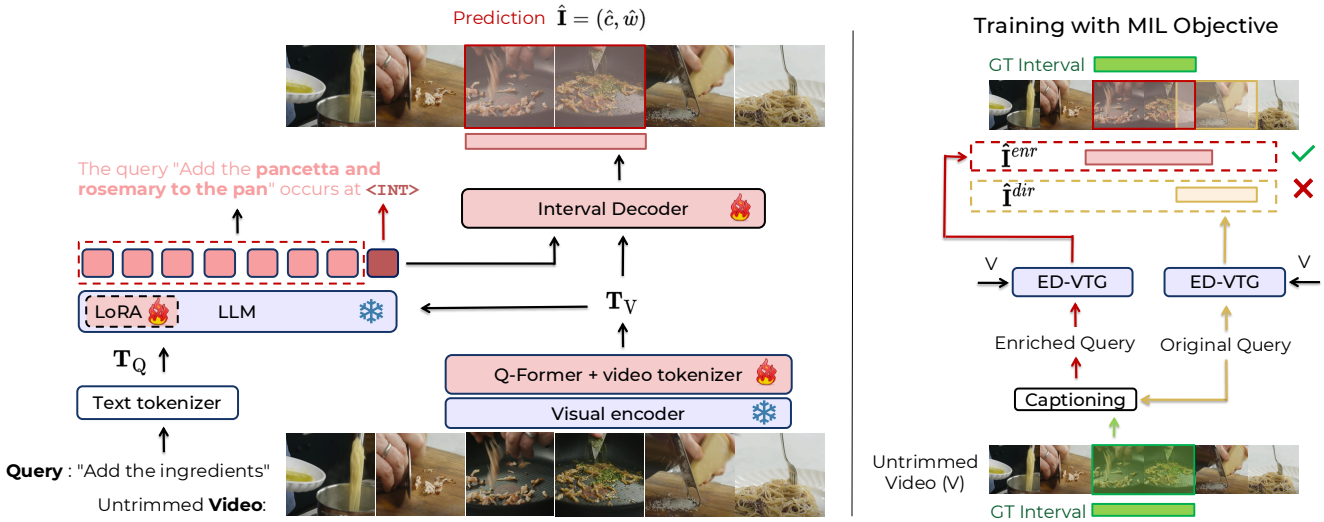

Figure 2. **Overview of the proposed Enrich and Detect framework**: *(left)* ED-VTG pipeline: Given an untrimmed video and a query Q to be grounded, the inputs are first tokenized into video tokens T_V and text tokens T_Q. The tokens are then fed into an LLM, which first generates an enriched query by, e.g., filling in any missing details and then emitting an interval token <INT>. The embedding of this special token is finally decoded into the predicted temporal interval via a lightweight interval decoder. In the example shown here, the vague input query is enriched into a more detailed one by our model which can be subsequently grounded more easily. *(right)* Training: ED-VTG is trained using ground-truth temporal intervals and pseudo-labels of enriched queries, generated by an external off-the-shelf captioning model [33], which – unlike our model at inference time – has access to the ground-truth intervals. For every sample during training, the proposed multiple instance learning (MIL) framework allows ED-VTG to assess both the original or the enriched queries and generate two sets of predictions, $\hat{I}^{enr}$, the interval predicted using the enriched query, and $\hat{I}^{dir}$ using the original query. Next, the model backpropagates using the better prediction (i.e., lower grounding loss). Hence, during training, ED-VTG dynamically learns to decide for which sample enrichment is necessary and, based on that, performs detection.

ing existing descriptions or providing new improved ones. LaViLa[109] uses rephraser and narrator models to improve the quality of the training data for video-text alignment. Augmentation during training is sufficient for that task, as the goal is to learn joint representations; for our grounding task however simply augmenting the training set, without enriching queries during inference is not as effective, as we empirically demonstrate.

Multiple instance learning. Multiple instance learning (MIL)[11] is a technique commonly used in weakly-supervised vision problems, when a collection of potential solutions is available but exact annotations are not. It has been successfully applied to a range of tasks, including classification [3], weakly supervised object detection[7, 9] and temporal action localization[57, 60].

3. Method

Given an untrimmed video $\mathcal{V}$ and a set of N associated textual queries $\mathcal{Q} = \{Q_1, \dots, Q_N\}$, temporal grounding aims to identify the corresponding temporal interval for each query. Formally, the output is a set of temporal intervals $\mathcal{I} = \{I_1, \dots, I_N\}$. For $N = 1$, the task takes the form of single-query temporal grounding, which we will simply refer to as STG. For clarity, we describe our approach in the STG setting, without loss of generality; the pipeline however readily extends to $N > 1$.

Our approach aims to tackle temporal grounding by leveraging a multimodal LLM that (i) transforms input queries into intermediate, enriched queries by adding missing details based on the video input, (ii) generates contextualized embeddings of each latent segment to enable temporal interval prediction, (iii) decodes these embeddings into concrete temporal boundaries.

In the following, we formally introduce our proposed ED-VTG model in Section 3.1, and proceed to describe how to train it with enriched query pseudo-labels within a MIL framework (Section 3.2).

3.1. Model

Our model (Figure 2) consists of three key modules: a vision encoder that extracts video representations, a LLM that jointly processes video and language, and a lightweight interval decoder that generates precise temporal boundaries.

Enrich. Given a single input query Q about a video V with T frames, the vision encoder represents the video as a sequence of R visual tokens $\mathbf{T}_V \in \mathbb{R}^{R \times D}$, where D is the token dimension. The tokenized video features $\mathbf{T}_V$ are fed along with the tokenized query $\mathbf{T}_Q$ to the LLM, which generates an enriched query $\hat{Q}^{enr} = \{\hat{y}_1, ..., \hat{y}_l, ...\hat{y}_{L^{enr}}\}$ one token at a time:

$$\hat{y}_l = \mathcal{F}_{\text{LLM}}(\hat{y}_{<l}, \mathbf{T}_V, \mathbf{T}_Q). \tag{1}$$

When the model is ready to ground the query, the LLM emits a new, special token <INT> to trigger interval prediction. In other words, the text prediction takes the form:

$$\hat{\mathbf{y}} = \text{“The query } \hat{Q}^{enr} \text{ occurs at <INT>”} \tag{2}$$

Detect. To detect the temporal interval corresponding to the enriched query $\hat{Q}^{enr}$, we introduce an interval decoder $\mathcal{F}_{\text{dec}}$ that predicts an interval $\hat{\mathbf{I}}$ parametrized by the center $\hat{c}$ and width $\hat{w}$ of the predicted interval. We selected this parameterization for its advantage in decoupling position from scale, as supported by literature in object detection [42, 68, 69, 114]. The interval decoder takes the form

$$\hat{\mathbf{I}} = (\hat{c}, \hat{w}) = \mathcal{F}_{\text{dec}}\left(\mathcal{G}(\mathbf{h}_{\text{int}}), \mathbf{T}_{\text{V}}\right), \tag{3}$$

where $\mathbf{h}_{int} \in \mathbb{R}^D$ is the hidden state of the LLM corresponding to the <INT> token and $\mathcal{G}$ is a linear projection layer. The decoder, functioning as the regression component of a temporal detector, consists of two transformer layers followed by a multi-layer perceptron (MLP), which processes a concatenation of its two inputs, $(\mathcal{G}(\mathbf{h}_{\text{int}}), \mathbf{T}_{\text{V}})$. It finally outputs the predicted interval $\hat{\mathbf{I}}$, grounding the input query $\mathcal{Q}$.

Notice that our formulation allows the quality of the enriched query $\hat{Q}^{enr}$ to directly influence the accuracy of the predicted interval $\hat{\mathbf{I}}$. This is because the generation of the interval token <INT> (and consequently its hidden state $\mathbf{h}_{\text{int}}$) is conditioned on the enriched query prediction. Establishing this *cascaded* dependency chain, i.e., $(V, Q) \rightarrow \hat{Q}^{enr}$ and $(V, \hat{Q}^{enr}) \rightarrow \hat{I}$, is the key idea of our approach.

3.2. Training

The model is trained end-to-end using two primary loss functions: a language modeling loss $\mathcal{L}_{\text{LM}}$ and a temporal grounding loss $\mathcal{L}_{\text{grnd}}$ that help supervise the “enrich” and “detect” aspects of our model, respectively.

Enrich. Given the target output text $\mathbf{y} = \{y_1, y_2, \ldots, y_L\}$, $\mathcal{L}_{\text{LM}}$ is calculated as the cross-entropy loss that evaluates the likelihood of $\mathbf{y}$ under the predicted probability distribution generated by the model:

$$\mathcal{L}_{\text{LM}} = -\sum_{t=1}^{T} \log P(y_l \mid \mathbf{y}_{<l}, \mathbf{T}_{\text{V}}, \mathbf{T}_{\mathcal{Q}}) \tag{4}$$

To provide a proper supervisory signal for query enrichment, we need the ground-truth pair of (Q, Q^{enr}). However, such datasets do not exist, nor is it practical to annotate a dataset solely for this purpose. Here, we capitalize on the tremendous amount of progress made in the video captioning literature [1, 78, 79], and use an off-the-shelf captioning model [33] to generate pseudo ground-truth Q^{enr} by refining the original query Q given its video V (see more details in Section 4.1).

Dataset	Domain	Tasks	Corpus	Eval. Protocol	# Train Samples	Avg Vid Length (s)	Avg Span Length (s)
DiDeMo [4]	Open	STG	PT	–	32.8K	54.57	6.49 (11.9%)
QuerYD [58]	Cooking	STG	PT	–	13.6K	158.78	7.68 (4.8%)
COIN [83]	Open	VPG	PT	–	7.5K	143.71	15.06 (10.5%)
HiREST [96]	Open	STG, VPG	PT	–	0.8K	208.03	44.50 (21.4%)
VITT† [23]	Open	VPG	PT	–	4.9K	287.17	–
YTTemporal [97]	Open	VPG	PT	–	28.8K	327.36	4.0 (1.2%)
CrossTask [118]	Procedural	AG	PT	–	2.7K	297.0	9.61 (3.2%)
VideoCC [55]	Open	STG	PT	–	45.0K	415.89	9.88 (2.3%)
Charades-STA [14]	Indoor	STG	FT, Eval	ZS, FT	12.4K	31.17	8.29 (26.6%)
Charades-CD-OOD [95]	Indoor	VPG	FT, Eval	FT	4.5K	30.60	7.90 (25.8%)
ANet-Captions [29]	Open	STG, VPG	FT, Eval	ZS, FT	9.5K	117.63	35.61 (30.3%)
TACoS [70]	Cooking	STG, VPG	FT, Eval	ZS, FT	9.8K	224.34	23.33 (10.4%)
YouCook2 [110]	Cooking	VPG	FT, Eval	FT	1.2K	311.41	20.07 (6.4%)
NExT-GQA° [90]	Open	QG	Eval	ZS	–	39.60	6.69 (16.9%)
HT-Step [2]	Cooking	AG	FT, Eval	FT	17.4K	393.89	14.88 (3.7%)

Table 1. **Dataset statistics, corresponding tasks, and evaluation protocol. The upper side** of the table represents datasets used for pre-training, resulting in a total of 136K samples. **The lower side** represents datasets used for fine-tuning and evaluation. We cover four different video grounding tasks: single-query temporal grounding (STG), video paragraph grounding (VPG), question grounding (QG), and article grounding (AG). QG is used only for evaluation to assess the model's generalization capability. Average interval lengths compared to the corresponding video durations are shown in brown, denoting the annotation granularity. †VITT contains single timestamp annotation instead of intervals. °NExT-GQA contains only evaluation split.

Detect. Given a target temporal interval $I = (c, w)$, the grounding loss $\mathcal{L}_{\text{grnd}}$ is computed as a combination of the L1 loss and the generalized Intersection over Union (gIoU) loss [73], applied on the predicted temporal interval $(\hat{c}, \hat{w})$:

$$\begin{aligned}\mathcal{L}_{\text{grnd}} = &\lambda_{\text{L1}}(|(\hat{c} - c| + |\hat{w} - w|) \\ &+ \lambda_{\text{gIoU}}\, \text{gIoU}((\hat{c}, \hat{w}), (c, w))\end{aligned} \tag{5}$$

3.2.1. MIL framework

A caveat with the pseudo-labeled enriched queries is that they can be noisy and include hallucinations; as a result, some of them may lead to a temporal interval prediction inferior to the original query. To address this, we propose considering *multiple* options for the target query (2 in the case of a single query scenario, Q and Q^{enr}), by adopting a multiple instance learning (MIL) framework [11]. This approach allows the model to choose between the enriched and the original input queries during training, depending on which version leads to a better temporal interval prediction. During inference, this means that our ED-VTG model can choose to either enrich the query if it misses important details, or decide to “carry over” the original query into $\hat{Q}^{enr}$ when it is concrete enough (in which case $Q = \hat{Q}^{enr}$).

Formally, to perform MIL, we collect two temporal interval predictions by running two forward passes with different LLM inputs (in a teacher-forcing fashion), *i.e.* $\mathbf{y}^{\text{dir}}$ that is formed using the original query $\mathcal{Q}$, and $\mathbf{y}^{\text{enr}}$ using the pseudo-labeled enriched query $\mathcal{Q}^{enr}$. We illustrate this schematically in Figure 2 (right). Recall that through the contextualized interval representation $\mathbf{h}_{\text{int}}$, the predicted interval depends on the LLM teacher-forced input. Hence, these inputs produce respective interval predictions, $\hat{\mathbf{I}}^{\text{dir}}$ and

Method	Generalist Model	# Train Samples	Eval.	Charades-STA				ActivityNet-Captions				TACoS			
				R@0.3	R@0.5	R@0.7	mIoU	R@0.3	R@0.5	R@0.7	mIoU	R@0.3	R@0.5	R@0.7	mIoU
UniVTG [40]	✗	4.2M	ZS	44.1	25.2	10.0	27.1	–	–	–	–	5.2	1.3	0.3	4.4
SeViLA [94]	✗	129M	ZS	–	–	–	–	31.6	19.0	10.1	23.0	–	–	–	–
PSVL [56]	✗	–	ZS	46.2	31.3	14.2	31.2	44.7	30.1	14.7	29.6	–	–	–	–
LT-ZVG [28]	✗	–	ZS	52.9	37.2	19.3	36.0	47.6	32.6	15.4	31.8	–	–	–	–
Video-LLaMA$^\diamond$ [103]	✓	2.7M	ZS	25.2	10.6	3.4	16.8	21.9	10.8	4.9	16.5	5.1	1.2	0.8	3.4
Video-ChatGPT$^\diamond$ [49]	✓	100K	ZS	27.2	6.2	1.9	19.7	19.5	10.6	4.8	14.2	6.3	1.7	1.0	4.3
Valley [47]	✓	100K	ZS	28.4	1.8	0.3	21.4	30.6	13.7	8.1	21.9	–	–	–	–
VideoChat2 [36]	✓	2M	ZS	38.0	14.3	3.8	24.6	40.8	27.8	9.3	27.9	–	–	–	–
Momenter [63]	✓	10M	ZS	42.6	26.6	11.6	28.5	42.9	23.0	12.4	29.3	–	–	–	–
VTimeLLM$^\diamond$ [21]	✓	170K	ZS	51.0	27.5	11.4	31.2	44.0	27.8	14.3	30.4	7.0	1.8	0.8	4.5
TimeChat$^\diamond$ [72]	✓	125K	ZS	–	32.2	13.4	–	–	–	–	–	6.8	2.1	0.8	4.7
HawkEye [87]	✓	715K	ZS	50.6	31.4	14.5	33.7	49.1	29.3	10.7	32.7	–	–	–	–
ChatVTG [66]	✓	100K	ZS	52.7	33.0	15.9	34.9	40.7	22.5	9.4	27.2	8.1	3.7	1.3	5.5
ED-VTG	✓	136K	ZS	**59.5**	**39.3**	**19.8**	**40.2**	**52.1**	**33.1**	**16.0**	**35.2**	**14.5**	**6.0**	**2.3**	**12.7**
$\Delta_{\text{Ours - HawkEye}}$	–	–	ZS	8.9 ↑	7.9 ↑	5.3 ↑	6.5 ↑	3.0 ↑	3.8 ↑	5.3 ↑	2.5 ↑	–	–	–	–
$\Delta_{\text{Ours - ChatVTG}}$	–	–	ZS	–	–	–	–	–	–	–	–	6.4 ↑	2.3 ↑	1.0 ↑	7.2 ↑

Table 2. **Zero-shot STG results on Charades, ActivityNet, and TACoS test splits.** For all three datasets, ED-VTG gains significant improvement over *all* existing methods, including task-specific, non-generalist models. We use **boldface** for the best and underline the second-best result for each metric, among the generalist models. $^\diamond$Official checkpoints are used for TACoS evaluation.

$\hat{\mathbf{I}}^{\text{enr}}$. We select the query version that results in the smallest grounding loss and use it to compute our overall training objective:

$$\mathcal{L} = \begin{cases} \lambda_{\text{LM}}\mathcal{L}^{\text{dir}}_{\text{LM}} + \lambda_{\text{grnd}}\mathcal{L}^{\text{dir}}_{\text{grnd}} & \text{if } \mathcal{L}^{\text{dir}}_{\text{grnd}} < \mathcal{L}^{\text{enr}}_{\text{grnd}} \\ \lambda_{\text{LM}}\mathcal{L}^{\text{enr}}_{\text{LM}} + \lambda_{\text{grnd}}\mathcal{L}^{\text{enr}}_{\text{grnd}} & \text{otherwise} \end{cases} \tag{6}$$

where $\mathcal{L}^{\text{dir}}_{\text{LM}}$ and $\mathcal{L}^{\text{enr}}_{\text{LM}}$ are the language losses for targets $\mathbf{y}^{\text{dir}}$ and $\mathbf{y}^{\text{enr}}$ respectively, and $\mathcal{L}^{\text{dir}}_{\text{grnd}}$ and $\mathcal{L}^{\text{enr}}_{\text{grnd}}$ the grounding losses for predictions $\hat{\mathbf{I}}^{\text{dir}}$ and $\hat{\mathbf{I}}^{\text{enr}}$ respectively. The hyperparameters λ_{LM} and λ_{grnd} are the relative weights of the language modeling and grounding losses.

4. Experiments

We design experiments to study three key questions related to our architecture and training framework: **Q1)** How does ED-VTG perform on various video grounding tasks in comparison to the current state-of-the-art? **Q2)** How beneficial is our query enrichment approach, within the MIL paradigm, as opposed to directly grounding the original queries? **Q3)** Does utilizing an interval decoder with specifically tailored grounding objectives offer advantages over predicting timestamps as raw text or special tokens?

4.1. Datasets

Table 1 summarizes the datasets that we used during the pre-training and fine-tuning stages. During pre-training, we use a total of 136K medium-to-long duration videos from 8 public datasets annotated with text queries and the corresponding intervals. As a preprocessing step, we collect pseudo-labels for enriched queries using an external captioning model [33]. In short, we take each video segment defined by the ground-truth intervals and prompt the captioning model to enrich the original query while preserving its meaning given the video segment. We provide full details of this process with the exact prompt used in the supplementary material.

4.2. Tasks

Table 1 also summarizes the tasks for which we use each dataset. Here we briefly describe these tasks, and highlight how our approach is applied to solve them.

Single-Query Temporal Grounding (STG) involves identifying a single time window in response to a single input language query (N=1).

Video Paragraph Grounding (VPG) involves grounding $N > 1$ sentences to N corresponding time windows. Our ED-VTG model can be trained to predict multiple enriched queries (one per original input query) interleaved with `<INT>` tokens that get separately decoded into intervals. Since there are multiple queries in the input, we run multiple forward passes through the LLM to perform MIL, wherein each pass selects a random number of queries to be enriched.

Question Grounding (QG) involves retrieving *evidence intervals* to answer questions, facilitating explainable video QA. As in STG, the input is a single query, and the output a single interval.

Article grounding (AG) is an extension of VPG where the model is given multiple queries as input, some of which may not be groundable in the video. Hence, the model must 1) identify which queries are groundable and 2) predict intervals for the groundable queries.

Note that, while we train our model jointly on multiple tasks, they are unified as a single task in the form of (Video, Text) → (Time intervals). Besides the differences in training data sources, we use the identical training objective (Eq. 6) during pre-training and fine-tuning stages.

4.3. Evaluation Protocol

Following the common practices in the literature [21, 72, 87], we evaluate ED-VTG in two primary evaluation protocols: (i) **Zero-shot (ZS)**, where the pre-trained model

Method	Generalist Model	# Train Samples	Eval.	Charades-STA				ActivityNet-Captions				TACoS			
				R@0.3	R@0.5	R@0.7	mIoU	R@0.3	R@0.5	R@0.7	mIoU	R@0.3	R@0.5	R@0.7	mIoU
VSLNet (C3D) [102]	✗	–	FT	64.3	47.3	30.2	45.2	63.2	43.2	26.2	43.2	29.6	24.3	20.0	24.1
MS-2D-TAN (I3D) [107]	✗	–	FT	–	56.6	36.2	–	62.1	45.5	28.3	–	42.0	33.6	22.1	–
Moment-DETR [32]	✗	236K	FT	65.8	52.1	30.6	45.5	–	–	–	–	38.0	24.7	12.0	25.5
UnLoc-B [91]	✗	650K	FT	–	58.1	35.4	–	–	48.0	29.7	–	–	–	–	–
MomentDiff [37]	✗	–	FT	–	55.6	32.4	–	–	–	–	–	46.6	28.9	12.4	30.4
LGI [54]	✗	–	FT	73.0	59.5	35.5	51.4	58.5	41.5	23.1	41.1	–	–	–	–
BAM-DETR [31]	✗	–	FT	72.9	60.0	39.4	52.3	–	–	–	–	56.7	41.5	26.8	39.3
InternVideo2* + CG-DETR [86]	✗	2.1M	FT	**79.7**	70.0	48.9	58.8	–	–	–	–	–	–	–	–
SG-DETR [18]	✗	–	FT	–	**71.1**	**52.8**	60.7	–	–	–	–	–	46.4	33.9	42.4
EMB (ELA) [24]	✗	–	FT	**79.7**	69.2	51.4	**62.2**	**73.7**	**58.7**	**40.7**	**56.2**	**63.3**	**52.5**	**37.0**	**48.4**
BLIP-2 (frames only) [34]	✓	129M	FT	–	43.3	32.6	–	–	25.8	9.7	–	–	–	–	–
VideoChat2 [36]	✓	2M	FT	–	–	–	–	55.5	34.7	17.7	38.9	–	–	–	–
TimeChat [72]	✓	125K	FT	–	46.7	23.7	–	–	–	–	–	27.7	15.1	6.4	18.4
HawkEye [87]	✓	715K	FT	72.5	58.3	28.8	49.3	55.9	34.7	17.9	39.1	–	–	–	–
VtimeLLM [21]	✓	170K	FT	–	–	–	–	–	–	–	–	26.8	14.4	6.1	18.0
ED-VTG	✓	136K	FT	**78.2**	**62.1**	**35.0**	**52.6**	**67.6**	**45.1**	**22.7**	**44.9**	**46.0**	**31.5**	**15.8**	**32.4**
$\Delta_{\text{Ours - HawkEye}}$	–	–	FT	5.7 ↑	3.8 ↑	6.2 ↑	3.3 ↑	11.7 ↑	10.4 ↑	4.8 ↑	5.8 ↑	–	–	–	–
$\Delta_{\text{Ours - VTimeLLM}}$	–	–	FT	–	–	–	–	–	–	–	–	19.2 ↑	17.1 ↑	9.7 ↑	14.4 ↑

Table 3. **Fine-tuned STG results on Charades, ActivityNet, and TACoS test splits.** For all datasets, ED-VTG achieves strong improvements over previous generalist models, and performs comparably to task-specific expert models. *Though InterVideo2 is a generalist model, it fine-tunes CG-DETR [52] head for grounding tasks, using the LLM as a feature extractor. For completeness, we report all competitive existing works, including task-specific SOTA specialist models (shown in gray), but directly compare ED-VTG only with generalist frameworks - using **boldface** for the best and underlining the second best results among generalist models (see discussion in Section 4.3).

is assessed directly without any fine-tuning on downstream datasets, and (ii) **Fine-tuned (FT)**, where the model undergoes additional training on specific tasks and datasets. We also present results without pre-training to assess its importance. For the STG and VPG tasks, following existing works [6, 21, 72, 76, 87], we report the mean intersection over union (mIoU) and Recall@1 for IoU$\geq m$ (R@m), with $m \in \{0.3, 0.5, 0.7\}$. For the QG task, we report intersection over prediction (IoP) in addition to IoU, following the NeXT-GQA [90] evaluation protocol. For AG, we adhere to the HT-Step [2] protocol and report article-grounding mAP scores across various IoU thresholds.

Comparison to non-generalist models. In the results section, we report all existing SOTA and competitive methods for a complete comparison, however we divide them into generalist and non-generalist (specialist) models. We note that specialist methods are heavily tailored to the task and in practice often overfit specific datasets which limits their transferability (as is evidenced by the zero-shot results in Table 2). We therefore focus the discussion around our method's comparison to other generalist models.

4.4. Implementation Details

We initialize the video encoder and LLM with the Video-LLaMA-7B [103] checkpoint, which is a similarly sized backbone used in existing LLM-based video grounding models [21, 63, 72, 87]. Video-LLaMA is trained on video captioning tasks with WebVid [5] and VideoChat [35]. The video encoder includes a ViT-G/14 from EVA-CLIP [81] as the frame feature extractor, followed by image and video QFormers. We initialize the decoder with random weights. We keep the ViT frozen, apply LoRA [20] with rank 32 to the LLM and fully tune the Q-Formers, decoder, and linear layers. We use LLaVA OneVision (OV) 72B [33] as the external captioner for obtaining enriched queries in the training sets. For the VPG task, we run four forward passes to perform MIL. We pre-train our model for 40 epochs with a batch size 256, using AdamW [45] with a peak learning rate of 5e-5 and a cosine scheduler [44] with a linear warmup for the first 20% steps. Pre-training takes 2 days on 16 V100 nodes (8 cards with 32G GPU memory each). Additional details on pre-training, fine-tuning, and task-specific instructions are provided in the supplementary material.

4.5. Results

Single-Query Temporal Grounding (STG). We start by comparing ED-VTG against the state-of-the-art across three different STG benchmarks, namely Charades-STA, ActivityNet-Captions, and TACoS, in a zero-shot evaluation setting. We show the results in Table 2. On Charades, ED-VTG achieves ZS scores of 59.5, 39.3, and 19.8 for R@0.3, R@0.5, and R@0.7, respectively, significantly outperforming all baseline models. Notably, ED-VTG surpasses Momenter [63] and HawkEye [87] by **11.4** and **6.2** absolute mIoU points despite these models being pre-trained with 100x and 6x more segment-level data, respectively. A similar trend is observed on ActivityNet and TACoS where our model achieves improvements of 2.5 and 7.2 mIoU points over the nearest LLM-based models in zero-shot setting. Notably, the TACoS dataset contains short, under-specified queries, longer input videos, and fine-grained interval annotations, which pose challenges for LLMs with a fixed number of input frames. The enriched query descriptions that ED-VTG generates enable it to more accurately align them to video frames and precisely retrieve the correct intervals.

On the fine-tuned STG evaluation, as shown in Table

Method	Generalist Model	Charades-CD-OOD R@0.3	R@0.5	mIoU
DepNet [6]	✗	45.6	27.6	29.3
DRN [98]	✗	40.5	30.4	–
STLG [46]	✗	48.3	30.4	–
SVPTR [26]	✗	50.3	28.5	32.1
SiamGTR [82]	✗	59.1	35.5	38.9
VTimeLLM† [21]	✓	53.2	34.0	35.1
TimeChat† [72]	✓	60.5	36.1	38.3
ED-VTG	✓	**70.7**	**47.3**	**45.0**
$\Delta_{\text{Ours - TimeChat}}$	–	10.2 ↑	11.2 ↑	6.7 ↑

(a) Results on **Charades-CD-OOD**.

Method	Generalist Model	ANet-Captions R@0.3	R@0.5	mIoU
CBLN [41]	✗	66.3	48.1	27.6
2D-TAN [105]	✗	59.5	44.5	–
3D-TPN [106]	✗	67.6	51.5	–
DepNet [6]	✗	72.8	55.9	–
SVPTR [26]	✗	**78.1**	**61.7**	**55.9**
VTimeLLM† [21]	✓	66.1	50.3	45.6
TimeChat† [72]	✓	67.9	51.5	47.0
ED-VTG	✓	**74.1**	**58.0**	**53.7**
$\Delta_{\text{Ours - TimeChat}}$	–	6.2 ↑	6.5 ↑	6.7 ↑

(b) Results on **ActivityNet**.

Method	Generalist Model	TACoS R@0.3	R@0.5	mIoU
CMIN [108]	✗	24.6	18.1	–
2D-TAN [105]	✗	37.3	25.3	–
3D-TPN [106]	✗	40.3	26.5	–
DepNet [6]	✗	41.3	27.2	–
SVPTR [26]	✗	**47.9**	**28.2**	**31.4**
VTimeLLM† [21]	✓	40.2	25.6	27.9
TimeChat† [72]	✓	39.5	25.6	27.8
ED-VTG	✓	**46.2**	**27.8**	**30.7**
$\Delta_{\text{Ours - TimeChat}}$	–	6.7 ↑	2.2 ↑	2.9 ↑

(c) Results on **TACoS**.

Method	Generalist Model	YouCook2 R@0.3	R@0.5	mIoU
DORi [75]	✗	43.4	30.5	30.5
DORi* [75]	✗	42.3	29.9	29.9
LocFormer [76]	✗	46.8	**31.3**	30.9
ExCL [16]	✗	26.6	16.2	18.9
TMLGA [74]	✗	34.8	23.1	24.4
VTimeLLM† [21]	✓	41.3	18.5	24.3
TimeChat† [72]	✓	40.9	19.0	26.6
ED-VTG	✓	**48.1**	**28.0**	**31.5**
$\Delta_{\text{Ours - TimeChat}}$	–	7.2 ↑	9.0 ↑	4.9 ↑

(d) Results on **YouCook2**.

Table 4. **Performance on VPG task on four different benchmarks: Charades-CD-OOD, ActivityNet-Captions, TACoS, YouCook2.** ED-VTG significantly improves over previous LLM-based models, and performs comparably to state-of-the-art specialist models. †We fine-tune VTimeLLM and TimeChat checkpoints for the VPG task. Dori* represents frozen text (BERT) encoder during fine-tuning.

Method	Generalist Model	NExT-GQA mIoP	IoP@0.3	IoP@0.5	mIoU	IoU@0.3	IoU@0.5
VGT [89]	✗	24.7	26.0	24.6	3.0	4.2	1.4
VIOLETv2 [13]	✗	23.6	25.1	23.3	3.1	4.3	1.3
Temp[CLIP] NG+ [90]	✗	25.7	31.4	25.5	12.1	17.5	8.9
FrozenBiLM NG+ [92]	✗	24.2	28.5	23.7	9.6	13.5	6.1
SeViLA [94]	✗	29.5	34.7	22.9	21.7	29.2	13.8
LLoVi 7B◇ [100]	✓	20.7	–	20.5	8.7	–	6.0
VideoStreaming◇ [64]	✓	32.2	–	31.0	19.3	–	13.3
LongRepo 7B◇ [27]	✓	20.3	–	20.0	8.7	–	6.0
DeVi [65]	✓	33.8	–	32.2	20.7	17.4	–
HawkEye [87]	✓	–	–	–	25.7	37.0	19.5
ED-VTG	✓	**34.7**	**45.1**	**33.5**	**26.6**	**39.5**	**19.8**
$\Delta_{\text{Ours - SeViLA}}$	–	5.2 ↑	10.4 ↑	10.6 ↑	4.9 ↑	10.3 ↑	6.0 ↑
$\Delta_{\text{Ours - HawkEye}}$	–	–	–	–	0.9 ↑	2.5 ↑	0.3 ↑

Table 5. **Performance on QG task on NeXT-GQA test split.** ED-VTG consistently achieves consistent improvements over the existing models across all metrics. ◇Results of LLoVi, LongRepo and VideoStreaming are from [64].

Method	Seen ↑ mAP@IoU @0.3	@0.5	@0.7	@[0.3-0.7]	Unseen ↑ mAP@IoU @0.3	@0.5	@0.7	@[0.3-0.7]
UMT [43]	15.7	8.7	3.2	9.1	9.4	4.9	1.7	5.3
MT+BCE [2, 50]	46.2	29.9	12.9	29.8	31.6	18.7	7.7	19.3
ActionFormer-T [101]	41.2	30.8	**18.3**	30.2	29.7	20.3	10.7	20.4
Timechat† [72]	45.3	29.0	14.4	29.0	30.7	17.8	7.5	18.7
ED-VTG	**48.9**	**31.5**	18.0	**32.5**	**33.0**	**21.2**	**11.1**	**21.6**
$\Delta_{\text{Ours - TimeChat}}$	3.6 ↑	2.5 ↑	3.6 ↑	3.5 ↑	2.3 ↑	3.4 ↑	3.6 ↑	2.9 ↑

Table 6. **Performance on AG task on HT-Step seen and unseen val split.** ED-VTG is the first LLM-based model to report results on for video grounding in the presence of negative, non-groundable queries and sets a new state of the art in terms of average mAP score across various IoU thresholds. †We fine-tune the official TimeChat checkpoint for AG task.

3, ED-VTG demonstrates an impressive gain of 5.7 and 11.7 points in R@0.3 over HawkEye on Charades and ActivityNet, respectively. Similarly on TACoS, ED-VTG surpasses all existing MLLMs by a considerable margin, e.g. 14.0 and 14.4 mIoU points over TimeChat and VTimeLLM. Furthermore, ED-VTG also beats many existing task-specific specialist models in all three benchmarks, significantly reducing the gap between specialist models and MLLMs for STG.

Video Paragraph Grounding (VPG). Next, we fine-tune ED-VTG for the VPG task, where the model processes multiple input queries in a temporal sequence. While specialist models have reported results in the past, no LLM-based models have previously addressed this challenging task. To establish a baseline for comparison, we fine-tune the officially released VTimeLLM and TimeChat pre-trained checkpoints. As shown in Table 4a, ED-VTG significantly outperforms both LLMs on the Charades-CD-OOD dataset, achieving an absolute gain of 9.9 and 6.7 mIoU points over VTimeLLM and TimeChat, respectively. Additionally, ED-VTG surpasses all specialist models on this dataset by a substantial margin, setting a new state-of-the-art.

We also evaluate VPG performance on three other benchmarks: ActivityNet-Captions, TACoS, and YouCook2, as presented in Tables 4b, 4c, and 4d. Consistent with the results on Charades, ED-VTG outperforms LLM-based models across all metrics on these datasets, with mIoU gains of 6.7, 2.9, and 4.9 over TimeChat on ActivityNet-Captions, TACoS, and YouCook2, respectively. ED-VTG also exceeds many existing specialist VPG models, demonstrating the effectiveness of our enriched queries and cascaded interval decoder.

Question Grounding (QG). To further assess the model's generalization capabilities, we conduct zero-shot evaluation on the held-out QG task. Our results on the NeXT-GQA test set are presented in Table 5. Notably, ED-VTG achieves state-of-the-art performance across all metrics, outperforming both specialist and LLM-based models by a significant margin. Specifically, ED-VTG surpasses the existing best baseline, HawkEye, by 2.5 points in terms of IoU@0.3 score, demonstrating its strong generalizability.

Article Grounding (AG). We assess ED-VTG on the AG on the HT-Step benchmark. Table 6 shows the fine-tuned performance on the AG task, where ED-VTG shows significant improvements over existing baselines. Notably, on the challenging *unseen* split, our model achieves the best results across all metrics, surpassing both the LLM and specialist models by a decent margin. The ability to handle non-groundable queries underscores ED-VTG's real-world applicability, as it does not always assume that the query is occurring in the input video.

4.6. Ablation Study

Effect of Enriched Queries. We examine the impact of query enrichment through a step-by-step ablation, as shown in Tables 7 and 8. Initially, we compare the effect of enrichment without MIL paradigm against direct grounding. As

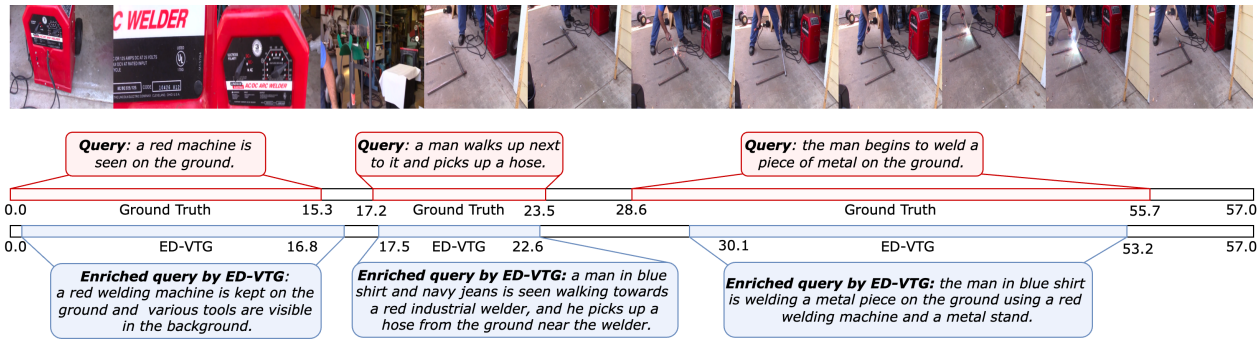

Figure 3. **Example of query enrichment and detection made by ED-VTG on video paragraph grounding (VPG) task from the ActivityNet-Captions [29] dataset.** In this specific sample, three different queries are enriched and localized together.

Training Paradigm	Charades-STA STG									ActivityNet-Captions STG								
	ZS			FT w/o PT			FT			ZS			FT w/o PT			FT		
	R@0.3	R@0.5	mIoU	R@0.3	R@0.5	mIoU	R@0.3	R@0.5	mIoU	R@0.3	R@0.5	mIoU	R@0.3	R@0.5	mIoU	R@0.3	R@0.5	mIoU
Detect	48.1	30.6	31.0	51.4	31.5	33.2	68.9	49.0	45.8	46.3	26.0	29.6	50.3	30.1	34.0	61.1	38.0	39.2
Enrich & Detect	58.1	37.3	37.7	60.1	37.0	38.4	75.1	56.6	49.7	50.7	29.5	33.4	56.3	35.5	37.8	65.5	43.4	43.8
Enrich & Detect w/ MIL	59.5	39.3	40.2	62.8	38.4	40.3	**78.2**	**62.1**	**52.6**	52.1	33.1	35.2	57.5	36.2	38.6	**67.6**	**45.1**	**44.9**

Table 7. **Ablation on the effect of enriched queries.** Our proposed enrich & detect framework significantly gains over directly grounding the input queries across different evaluation settings. Introducing the MIL framework further improve the performance. FT w/o PT refers directly fine-tuning on respective datasets, without performing pre-training.

Training Paradigm	Charades-STA STG			ANet-Captions STG		
	R@0.3	R@0.5	mIoU	R@0.3	R@0.5	mIoU
Detect	51.4	31.5	33.2	50.3	30.1	34.0
Offline Enrich + Detect	51.7	31.5	33.4	49.8	29.9	33.7
Enrich & Detect	**60.1**	**37.0**	**38.4**	**56.3**	**35.5**	**37.8**

Table 8. **Ablation on enrichment as a training pre-processing step.** The two step enrich & detect framework is more helpful since the trained model learn to perform autonomous enrichment during evaluation. Reported results are in FT w/o PT setting.

Decoder	Objectives			Charades-STA STG			ANet-Captions STG		
	LM	L1	gIoU	R@0.3	R@0.5	mIoU	R@0.3	R@0.5	mIoU
–	✓	–	–	54.2	33.2	34.1	51.0	31.6	35.5
✓	✓	✓	–	58.5	36.0	37.0	55.8	34.4	37.1
✓	✓	–	✓	58.9	36.2	37.1	55.8	34.7	37.3
✓	✓	✓	✓	**60.1**	**37.0**	**38.4**	**56.3**	**35.5**	**37.8**

Table 9. **Ablation study on different training objectives.** Training the model only using the LM loss (without the decoder) leads to significant performance drop. Results are in FT w/o PT setting.

indicated in the first two rows of Table 7, enriched queries lead to significant improvements on the Charades-STA and ActivityNet-Captions STG benchmarks. In the zero-shot (ZS) setting, enrichment results in gains of 6.7 and 3.8 mIoU points on these datasets, respectively. Introducing the MIL paradigm further enhances performance, adding 2.5 and 1.8 mIoU points. Similar improvements are observed in other evaluation settings, highlighting the substantial effectiveness of query enrichment within the MIL framework.

Additionally, we investigate the effect of offline enrichment in Table 8. In this scenario, instead of using a two-step grounding process, we enrich the queries as a training enrichment step, and the model is then directly provided with these enriched queries as input and asked to directly perform grounding. However, we find that offline enrichment is not advantageous, primarily due to the lack of enrichment during evaluation. In contrast, our two-step grounding approach allows the trained model to learn how to enrich queries and improve them autonomously when necessary, resulting in significant performance gains.

Training Objectives. We ablate different training objectives on the Charades and ActivityNet STG benchmarks, as shown in Table 9. The decoder achieves optimal performance when both L1 and gIoU objectives are used together; omitting either one slightly reduces the scores.

4.7. Qualitative Results and Error Analysis

Figure 3 visualizes a VPG sample from the ActivityNet-Captions dataset where ED-VTG meaningfully enriches the three input queries, then proceeds to precisely ground them. Please refer to supplementary for more qualative results, including intuitive demonstrations of the flexibility in choosing between enrichment or direct detection, and comparison with TimeChat basline. We also present there some interesting failure cases, such as the ones involving small and obscured objects in long input videos.

5. Conclusion

In this paper, we presented ED-VTG, a novel method for fine-grained video temporal grounding using multi-modal LLMs. By enhancing queries with additional details, utilizing a lightweight decoder and trained in a multiple-instance framework, ED-VTG accurately locates temporal boundaries in videos. Our experiments show that our method outperforms existing LLM-based works and is competitive with specialized models, especially in zero-shot settings, setting a new, strong benchmark for video grounding tasks.

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